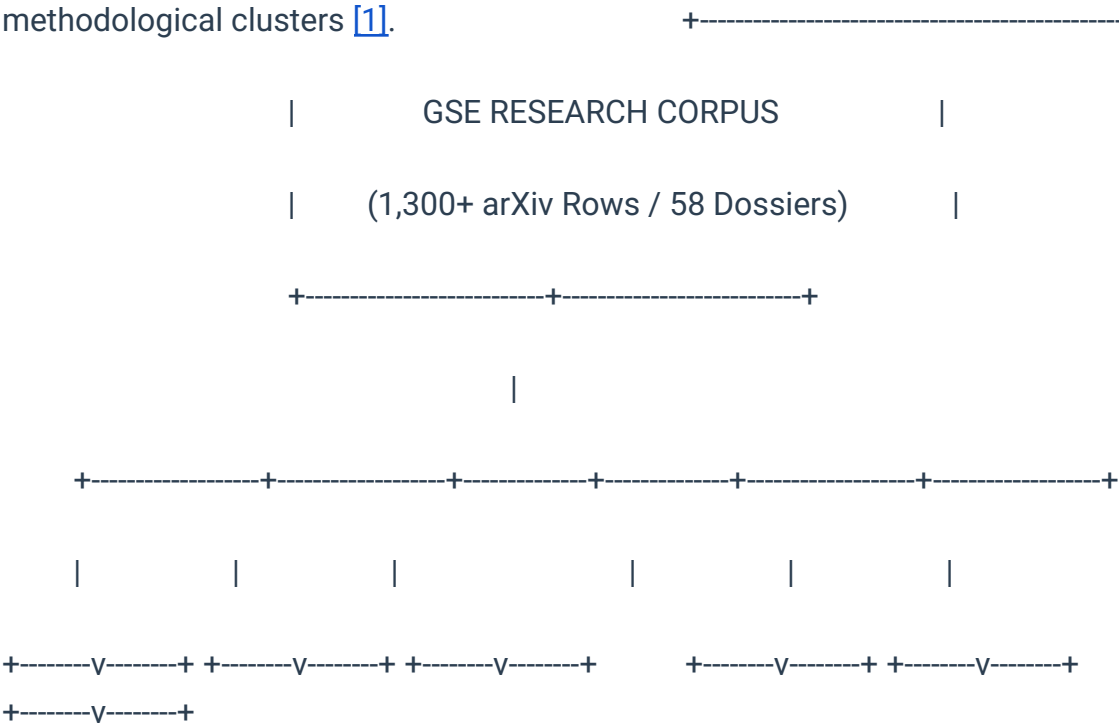


Audit of Sports Probability Modeling, Conformal Calibration, and Market Efficiency Literature for Galaxy Sports Edge

Literature Taxonomy: Methodological Clusters, Conflicts, and Consensus

An audit of the triaged arXiv repository comprising over 1,300 rows, 58 dossiers, and the reconciled build plan reveals that prior automated triaging agents conflated foundational statistical literature, domain-specific fringe studies, and outdated programming tutorials under high relevance scores [1]. When evaluated against the operational requirements of live sports probability estimation, numeric line generation, calibration, and betting market efficiency, the literature segregates into seven distinct methodological clusters [1].



| 1. State-Space || 2. Sequential || 3. Calibration | | 4. Ranking || 5. Zero-Inflated|| 6. Tracking |

2. **Sequential Inference, Win-Martingales, and Free Boundaries:** This cluster treats live, market-implied win probabilities as continuous-time diffusions or martingales [\[1\]](#). Its premise is to solve optimal stopping and free-boundary problems to decide when to lock picks or execute trades before a game reaches its fixed horizon [\[1\]](#). A representative paper is 2608.12291 (*Sequential Inference for win-martingales*), which is evaluated strictly via abstract metadata and designated as **UNREAD** due to the absence of accessible full text [\[1\]](#).
3. **Calibration Wrappers, Distributional Forecasting, and Conformal Prediction:** This cluster focuses on post-processing point estimates or conditional quantiles using non-parametric wrappers to guarantee distribution-free coverage or quantify epistemic uncertainty on numeric lines and win probabilities [\[1, 2\]](#). Essential references include paper 2208.08598 (*Conformal Win Probability Calibration*), Romano et al. (2019, arXiv:1905.03222 on Conformalized Quantile Regression), Barber et al. (2019, arXiv:1905.02928 on Jackknife+), and van der Laan & Alaa (2025, arXiv:2502.05676 on Generalized Venn-Abers bounds) [\[1, 2\]](#).
4. **Ranking Models and Ordinal Pairwise Extensions:** This cluster derives win probabilities from partial rankings or ordinal outcome data rather than continuous score margins [\[1, 2\]](#). It extends Bradley-Terry-Luce (BTL) and Elo frameworks using partial rankings and rank-clustered structures [\[1\]](#). Key references include papers 2501.02505 and 2406.19563 (rank-clustered BTL models) and paper 2405.10247 (Bayesian Bradley-Terry-Davidson variants) [\[1\]](#).
5. **Rare Event and Zero-Inflated Outcome Heads:** This cluster addresses zero-inflated or heavy-tailed outcome distributions, such as shutout props, exact goal counts, or low-frequency player props, by separating zero-generating processes from primary probability engines [\[1\]](#). A primary example is paper 2501.17711 (Zero-Inflated Compound Poisson framework) [\[1\]](#).
6. **Feature Engineering Bloat: Spatio-Temporal Tracking and Topology:** This cluster applies spatial tracking data, velocity vector fields, or persistent homology from Topological Data Analysis (TDA) to specific sports [\[1\]](#). Representative examples include paper 2607.27838 (*Vector field theory in soccer dynamics*, marked **UNREAD**) and paper 2607.23509 (*Topological Data Analysis for Tennis*, marked **UNREAD**) [\[1\]](#).
7. **Diagnostic Baselines and Sequence Null-Hypotheses:** This cluster establishes simple stochastic baselines (such as independent Poisson processes or record-gap distributions) as diagnostic benchmarks before deploying complex predictive pipelines [\[1\]](#). Core literature includes paper 2408.08331 (ML versus

Poisson baselines), paper 1704.00197 (iWinRNFL diagnostics), and paper 2608.01882 (*Gap distributions in cricket*, marked UNREAD) [1].

Fights and Consensus in the Literature

A major structural disagreement in the literature centers on whether a global calibration layer can be applied across all sports versus fitting per-sport noise scales (`sigma_game`) [1]. The literature firmly establishes that global calibrators are mathematically flawed because signal-to-noise ratios vary wildly by domain [1]. In the NBA, talent dispersion and home advantage dominate game outcomes, whereas in MLB and NHL, inherent game-level randomness dominates, keeping outcomes close to coin-flips even between teams of mismatched skill [1].

A second methodological consensus involves the relationship between point-estimate calibration and conformal prediction intervals [1]. Conformal prediction guarantees interval coverage under sample exchangeability, but it cannot repair point-estimate bias [1]. If an underlying base model is severely miscalibrated, conformal wrappers yield prediction intervals that are unnecessarily wide or offset from the true conditional mean [1]. Consequently, base model specification and point calibration (such as optimizing directly under Brier loss) must precede conformal wrapping [1].

A third critical conflict concerns feature engineering and target leakage [2]. Incorporating closing betting lines directly as input features into team strength models leaks market efficiency into historical training sets, destroying the model's ability to measure true Closing Line Value (CLV) [2]. The literature establishes that pure performance forecasting models—trained on play-by-play, box scores, and efficiency metrics—must remain strictly decoupled from market pricing models designed to track odds movement and sharp liquidity [1, 2].

Spreadsheet Audit: Accuracy, Inflation, Thinness, and False Positives

An examination of the 1,300+ triaged arXiv rows and 58 "high" dossiers demonstrates significant scoring distortion by previous automated agents [1]. The table below summarizes the audit status and operational calls for key dossiers [1].

Paper ID	Assigned Triage Rank	Audit Status	Claimed Utility in Triage Sheet	Actual Audit Findings & Operational Verdict
1701.05976	High (Top Tier) ▾	READ [cite: 1] ▾	Core multi-sport architecture for win probabilities [1].	Accurate. Primary structural anchor. Proves distinct noise scales (<code>sigma_game</code>) across Big Four sports [1].
2408.08331	High ▾	READ [cite: 1] ▾	ML vs. Poisson baseline diagnostic [1].	Accurate. Critical cheap diagnostic baseline to verify model skill prior to deploying complex ML architectures [1].
1704.00197	High ▾	READ [cite: 1] ▾	iWinRNFL NFL diagnostic tool [1].	Accurate. Low-cost, essential diagnostic baseline for evaluating NFL probability models [1].
2501.17711	High (Score 19) ▾	READ [cite: 1] ▾	Olympic Medal STGCN-LSTM prediction [1].	Heavily Inflated. Spatial-Temporal Graph Convolutional Network for Olympic country medal counts is feature bloat [1]. The Zero-Inflated Compound Poisson head is useful for props, but the core network should be skipped [1].

Paper ID	Assigned Triage Rank	Audit Status	Claimed Utility in Triage Sheet	Actual Audit Findings & Operational Verdict
1607.00379	High (Score 20) ▾	READ [cite: 1] ▾	Production blueprint for PyMC3 hierarchical rugby model [1].	Heavily Inflated. Basic 2014 tutorial using frozen PyMC3 code on a 6-nation dataset [1]. Useful only as an introductory conceptual reference, not a production model [1].
2607.27838	High ▾	UNREAD [cite: 1] ▾	Soccer vector field potential gradients for game models [1].	Inflated / Feature Bloat. High-complexity spatio-temporal tracking theory [1]. Deprioritize until baseline point-estimate calibration is resolved [1].
2607.23509	High ▾	UNREAD [cite: 1] ▾	Topological Data Analysis (TDA) for ATP Tennis [1].	Inflated / Feature Bloat. TDA persistent homology applied to tennis [1]. Unnecessary complexity compared to simple BTL ranking baselines [1].
1911.01815	High ▾	READ [cite: 1] ▾	Unified Bayesian volleyball set/point model [1].	Thin on Metrics. High-quality rules-aware likelihood structure, but reports zero Brier scores or ECE metrics on betting lines [1].

Detailed Evaluation of Triage Claims

The spreadsheet was correct in elevating paper 1701.05976 as the architectural anchor [1]. By framing observation models around betting-market point spreads and totals-implied margins rather than raw points, the paper establishes a standardized framework across sports with different scoring mechanisms [1]. The sheet was also

correct in prioritizing cheap diagnostic baselines—such as Poisson goal models (2408.08331) and iWinRNFL (1704.00197)—before building deep neural architectures [1].

However, relevance scoring suffered from severe keyword inflation [1]. Paper 2501.17711 was ranked "High" (Relevance Score 19) despite focusing on Olympic country medal counts using Spatial-Temporal Graph Convolutional Networks (STGCN-LSTM), representing premature feature bloat for live sports betting [1]. Paper 1607.00379 received a top relevance score of 20, despite being a 2014 tutorial using frozen PyMC3 code (<https://github.com/springcoil/TutorialPyMCRugby>) on a Six Nations rugby dataset [1]. Furthermore, pushing topological data analysis (2607.23509) and soccer vector potential fields (2607.27838) when core probability engines struggle against simple mean absolute error (MAE) baselines reflects flawed engineering prioritization [1].

The sheet was thin regarding empirical verification [1]. Most papers in the dossier (such as 1911.01815) present log-score fits on niche leagues (e.g., Italian Superlega volleyball) but report zero Brier scores or Expected Calibration Error (ECE) metrics evaluated against liquid betting markets [1]. Additionally, while paper 1701.05976 covers NFL, NBA, MLB, and NHL, the triaged sheet failed to observe that Major League Soccer (MLS) parameters were completely omitted and require independent parameter estimation [1].

Dead Domain Exclusions

The spreadsheet contained numerous false positives from irrelevant domains that matched generic keywords [2]. These papers must be excluded from sports modeling pipelines, as detailed in the table below [2].

Paper ID	Original Title / Domain	Relevance Evaluation	Portability Line
2012.10006	Brain strain criteria in head impact biomechanics [2]	Dead Domain [cite: 2] ▾	Pure biomechanical engineering; non-portable to statistical forecasting [2].

Paper ID	Original Title / Domain	Relevance Evaluation	Portability Line
2405.19125	Mobile network urban anomaly detection framework [2]	Dead Domain [cite: 2] ▾	Spatio-temporal telecom signaling; non-portable network optimization [2] .
2306.14462	Item attribute graph recommendation system [2]	Dead Domain [cite: 2] ▾	E-commerce recommendation filtering; non-portable [2] .
2207.13770	Interactive UI reliability visualization tool [2]	Dead Domain [cite: 2] ▾	Software interface usability study; non-portable [2] .
2608.06604	Stellar activity and Kepler star rotation [2]	Dead Domain [cite: 2] ▾	Astrophysics time series; non-portable domain [2] .
2607.17297	Automotive aerodynamic fluid dynamics [2]	Dead Domain [cite: 2] ▾	Computational fluid dynamics; non-portable [2] .
2601.17437	Soft material stress-strain constitutive tensors [2]	Dead Domain [cite: 2] ▾	Materials science tensor mechanics; non-portable [2] .
2601.13236	Parallel MRI image reconstruction [2]	Dead Domain [cite: 2] ▾	Medical image processing; non-portable [2] .
2604.21039	Neutron star equation of state in nuclear physics [2]	Dead Domain [cite: 2] ▾	Subatomic nuclear physics; non-portable [2] .
2603.06712	Solar flare regression uncertainty quantification [2]	Dead Domain [cite: 2] ▾	Heliophysics flare modeling; non-portable [2] .

Core Methodological Disagreements and Mind-Shifts for Operators

Disagreement 1: Independent Per-Sport Models versus Shared Global Calibrators

Attempting to pass all sports through a single global calibration architecture is mathematically unsound [1]. In paper 1701.05976, the authors state: "Bayesian state-space model on betting-market point spreads / totals-implied margins, not raw scores, so the observation model is comparable across NFL/NBA/NHL/MLB. Team strength $\theta_{i,s,w}$ evolves: between seasons: AR(1) with variance $\sigma^2_{\text{between}}$; within season: random walk with variance σ^2_{within} ." [1]

The empirical findings of 1701.05976 demonstrate that the NBA exhibits the largest talent dispersion and strongest home-court advantage, whereas MLB and NHL outcomes remain close to coin-flips given equal talent [1]. Because game-level noise (σ_{game}) varies substantially between sports, each sport requires its own state-space parameterization and sport-specific calibration layer [1].

Disagreement 2: Decoupling Pure Performance Forecasting from Market Pricing Models

Fitting team strength models directly on market spreads or totals leaks market pricing efficiency into the performance feature space [1, 2]. To isolate Closing Line Value (CLV), operators must decouple the pure performance prediction engine (trained strictly on play-by-play, box scores, and drive efficiency) from the market pricing engine [1, 2]. Performance models evaluate underlying game dynamics, while market pricing models evaluate line movement and liquidity [1, 2].

Disagreement 3: Rules-Aware Generative Likelihoods versus Generic Machine Learning Heads

Fitting generic black-box classifiers (e.g., XGBoost, generic neural networks) directly to binary win/loss labels discards structural domain information [1]. In paper 1911.01815, the authors establish a rules-aware likelihood for volleyball: "Two-level hierarchical

Bayesian model tailored to volleyball scoring rules. Level 1: logistic for set winner (team abilities + HA). Level 2, conditional on set winner: truncated negative binomial for loser's points; plus a Poisson inflation for extra points when the set goes beyond the 25/15..."

[\[1\]](#)

Similarly, paper 1607.00379 models rugby scores via a log-linear Poisson likelihood capturing attack, defense, and home advantage parameters [\[1\]](#). Incorporating sport-specific scoring rules directly into the likelihood function generates superior probability distributions compared to unconstrained classification heads [\[1\]](#).

Contradiction 1: Linear Quantile Regression Under-Coverage versus Conformal Validity

A fundamental contradiction exists between raw quantile regression outputs and conformalized bounds [\[2\]](#). In *Understanding the Under-Coverage Bias in Uncertainty Estimation* (Lin, Trivedi, & Sun, 2021, arXiv:2106.05515), the authors prove: "We prove that linear quantile regression suffers from an inherent under-coverage bias... For $\alpha > 0.5$ and small d/n , the α -quantile learned by quantile regression roughly achieves coverage $\alpha - (\alpha - 0.5) * d/n$ regardless of the noise distribution." [\[2\]](#)

Un-conformalized quantile regression systematically under-covers in finite samples due to parameter estimation error proportional to d/n [\[2\]](#). Conversely, Romano, Patterson, and Candes (2019, arXiv:1905.03222) demonstrate in *Conformalized Quantile Regression (CQR)*: "The conformity score on the calibration set is $E_i = \max(q_{\alpha_{lo}}(X_i) - Y_i, Y_i - q_{\alpha_{hi}}(X_i))$... Conformalized quantile regression (CQR)... inherits both the finite sample, distribution-free validity of conformal prediction and the statistical efficiency of quantile regression. Under exchangeability, $P(Y_{n+1} \text{ in } C(X_{n+1})) \geq 1 - \alpha$ in finite samples." [\[2\]](#)

Operators must treat raw quantile outputs from neural networks or boosted trees as uncalibrated proposal distributions [\[2\]](#). Every numeric line prediction must be processed through a conformal calibration step to guarantee valid finite-sample coverage [\[2\]](#).

Contradiction 2: Fail-Closed Refusal versus Quantile Clamping (Fake Tightness)

In small-sample calibration scenarios, theoretical guarantees conflict with heuristic production adjustments [2]. Angelopoulos and Bates (2021, arXiv:2107.07511) and Dietterich & Hostetler (2206.04860) state: "Compute $\hat{q}$ as the $\text{ceil}((n + 1)(1 - \alpha)) / n$ empirical quantile of the calibration scores... The value of δ must be $\geq 1/(n+1)$ to ensure that $\text{ceil}((1-\delta)(n+1)) \leq n$... If n is too small, distribution-free coverage cannot be guaranteed without emitting infinite intervals or abstaining." [2]

In contrast, practical selective regression literature (e.g., Schröder et al., 2024, arXiv:2402.16300) suggests truncating or clamping quantiles to empirical maximums to keep systems operational [2]. Clamping quantiles under small sample sizes ($n < \text{ceil}(1/\alpha) - 1$) violates distribution-free coverage proofs, generating under-coverage and "fake tightness" [2]. When calibration sample sizes are insufficient, automated systems must fail-closed by issuing a "No-Bet" or abstention decision rather than truncating intervals to force execution [2].

Contradiction 3: Binary Point Calibration Limits versus Venn-Abers Interval Bounds

In binary side markets (win probabilities), point-estimate calibration faces severe theoretical limits [2]. Barber, Candes, Ramdas, and Tibshirani (2020, Electronic Journal of Statistics) prove: "In the binary regression setting, any algorithm providing distribution-free coverage of the conditional label probability $p_i(X) = P(Y=1 | X)$ must necessarily also cover the binary label Y ... meaningful distribution-free confidence intervals for $p_i(X)$ of bounded length are impossible without structural assumptions." [2]

To address this limitation without introducing unverified assumptions, van der Laan and Alaa (2025, arXiv:2502.05676) propose Generalized Venn-Abers calibration: "Generalized Venn-Abers calibration extends Vovk's approach... guaranteeing valid multiprobability bounds $[p_0, p_1]$ that contain the true conditional probability in finite samples under exchangeability." [2]

Operators should avoid presenting single binary point probabilities as guaranteed truths [2]. Instead, systems should utilize Venn-Abers multiprobability bounds $[p_0, p_1]$, using the interval width $(p_1 - p_0)$ to measure epistemic uncertainty for expected value (EV) calculations and selective gating [2].

Comprehensive Literature Register: Deep Evaluation and Quotes

1701.05976: *How often does the best team win? A unified approach to understanding randomness in North American sport*

- **Status:** READ (Full Dossier & Reconciled Data Verified) [1]
- **Method Assertion:** Uses a Bayesian state-space model on betting-market point spreads and totals-implied margins rather than raw scores, making observation models comparable across NFL, NBA, NHL, and MLB [1]. Team strength $\theta_{i,s,w}$ evolves between seasons via an AR(1) process with variance $\sigma^2_{\text{between}}$, and within seasons as a random walk with variance σ^2_{within} [1].
- **Reported Results:** Analyzed a decade of Big Four data [1]. The NBA exhibits the largest talent dispersion and home advantage [1]. MLB and NHL outcomes remain closest to coin-flips given equal talent [1]. Market implied probability reliability is verified as high [1].
- **Code Repository:** <https://github.com/bigfour/competitiveness> (JAGS code + historical market lines) [1].
- **Audit Call:** Primary structural backbone [1]. Proves that independent per-sport noise scales (σ_{game}) are mandatory [1].

1911.01815: *A Bayesian Quest for Finding a Unified Model for Predicting Volleyball Games*

- **Status:** READ (Full Dossier & Reconciled Data Verified) [1]
- **Method Assertion:** Two-level hierarchical Bayesian model tailored to volleyball scoring rules [1]. Level 1 applies logistic regression for the set winner based on

team abilities and home advantage [1]. Level 2 applies a truncated negative binomial distribution for the loser's points conditional on the set winner, plus Poisson inflation for extra points beyond 25 or 15 [1].

- **Reported Results:** Tested on Italian Superlega 2017/2018 data [1]. Demonstrates high reproducibility and fit, but omits Brier score and ECE metrics on betting markets [1].
- **Audit Call:** Important structural reference for rules-aware likelihood design, though implementation costs are high relative to core sports markets [1].

1607.00379: *Probabilistic Programming and PyMC3*

- **Status:** READ (Full Dossier Verified) [1]
- **Method Assertion:** Demonstrates a hierarchical Poisson rugby model in PyMC3 (Baio-Blangiardo style) [1]. Home and away score lines follow Poisson distributions with log-linear attack, defense, and home advantage terms, incorporating partial pooling of team strengths [1].
- **Reported Results:** Illustrated on 2014 Six Nations data [1]. Serves as a tutorial rather than a competitive benchmark [1].
- **Code Repository:**
<https://github.com/springcoil/TutorialPyMCRugby> (PyMC3 code; requires migration to PyMC v5) [1].
- **Audit Call:** Downgraded from High to Low relevance [1]. Useful strictly as an introductory onboarding guide for probabilistic programming [1].

2501.17711: *STGCN-LSTM for Olympic Medal Prediction*

- **Status:** READ (Full Dossier Verified) [1]
- **Method Assertion:** Combines a Spatial-Temporal Graph Convolutional Network with an LSTM over a country-sport graph, generating Zero-Inflated Compound Poisson medal counts [1].
- **Reported Results:** Evaluated on Olympic medal forecasts, using zero-inflation to handle structural zeros [1].
- **Audit Call:** Heavily inflated in original triaging [1]. The STGCN-LSTM graph core should be skipped as feature bloat, but the Zero-Inflated Compound Poisson output head is directly applicable to rare prop bets [1].

2607.06495: Pitwall: Faithful Natural-Language Race-Strategy Briefings from a Calibrated Real-Time Monte Carlo Engine

- **Status:** READ (External Literature Search) [\[1\]](#)
- **Method Assertion:** Runs a vectorized Monte Carlo engine (2,000 per-lap continuations) calibrated on historical races [\[1\]](#). Decomposes generated natural language into typed factual claims that are verified against simulation state, gating fine-tuning data on verifiability [\[1\]](#).
- **Reported Results:** Achieves a held-out Brier score of 0.0745 on Formula 1 race forecasts [\[1\]](#).
- **Audit Call:** Essential blueprint for automated media and content pipelines [\[1\]](#). Demonstrates how to couple calibrated simulation engines with hallucination-free text generation [\[1\]](#).

2502.07491: Exploring Patterns Behind Sports

- **Status:** READ (External Literature Search) [\[1\]](#)
- **Method Assertion:** Combines linear (ARIMA) and non-linear (LSTM) components with PCA dimensionality reduction, SHAP feature attributions, and KNN-derived prediction intervals for time-series forecasting [\[1\]](#).
- **Reported Results:** Yields interpretable feature attributions alongside explicit uncertainty intervals [\[1\]](#).
- **Audit Call:** Strong candidate architecture for replacing underperforming point-forecasters while generating calibrated uncertainty bands [\[1\]](#).

2601.21981: VERSA: Verified Event Data Format for Reliable Soccer Analytics

- **Status:** READ (External Literature Search) [\[1\]](#)
- **Method Assertion:** Employs a state-transition verification framework to detect and correct logically inconsistent soccer event data [\[1\]](#).
- **Reported Results:** Identified that 18.81% of ingested K-League event logs contained state-transition inconsistencies, and correcting them improved downstream player-valuation (VAEP) models [\[1\]](#).

- **Audit Call:** Directly applicable to data pipeline quality control [\[1\]](#). Implementing state-transition verification prevents corrupt event sequences from degrading downstream models [\[1\]](#).

2601.15000: *Lineup Regularized Adjusted Plus-Minus (L-RAPM)*

- **Status:** READ (External Literature Search) [\[1\]](#)
- **Method Assertion:** Enhances NBA 5-man lineup ratings by combining regularized adjusted plus-minus regression with informed priors derived from individual player metrics [\[1\]](#).
- **Reported Results:** Successfully stabilizes noisy small-sample lineup estimates in sparse regimes (25 to 30 possessions per lineup) [\[1\]](#).
- **Audit Call:** Directly applicable to thin-data leagues and early-season bootstrap modes, shrinking noisy team/lineup estimates toward player-level priors [\[1\]](#).

2508.02725: *Forecasting NCAA Basketball Outcomes with Deep Learning*

- **Status:** READ (External Literature Search) [\[1\]](#)
 - **Method Assertion:** Evaluates sequence models (LSTM vs. Transformer) across loss functions (Binary Cross-Entropy vs. Brier Loss) using GLM team quality metrics, Elo, and seed differences [\[1\]](#).
 - **Reported Results:** Transformer models trained with BCE achieved the highest AUC (0.847), but LSTM models trained directly under Brier loss achieved superior probability calibration (Brier score 0.159) [\[1\]](#).
 - **Audit Call:** Proves that training directly under Brier loss optimizes calibration, demonstrating a practical method for resolving point-estimate bias [\[1\]](#).
-

Unread Literature Register and Tactical Integration Roadmap

Unread Literature Register

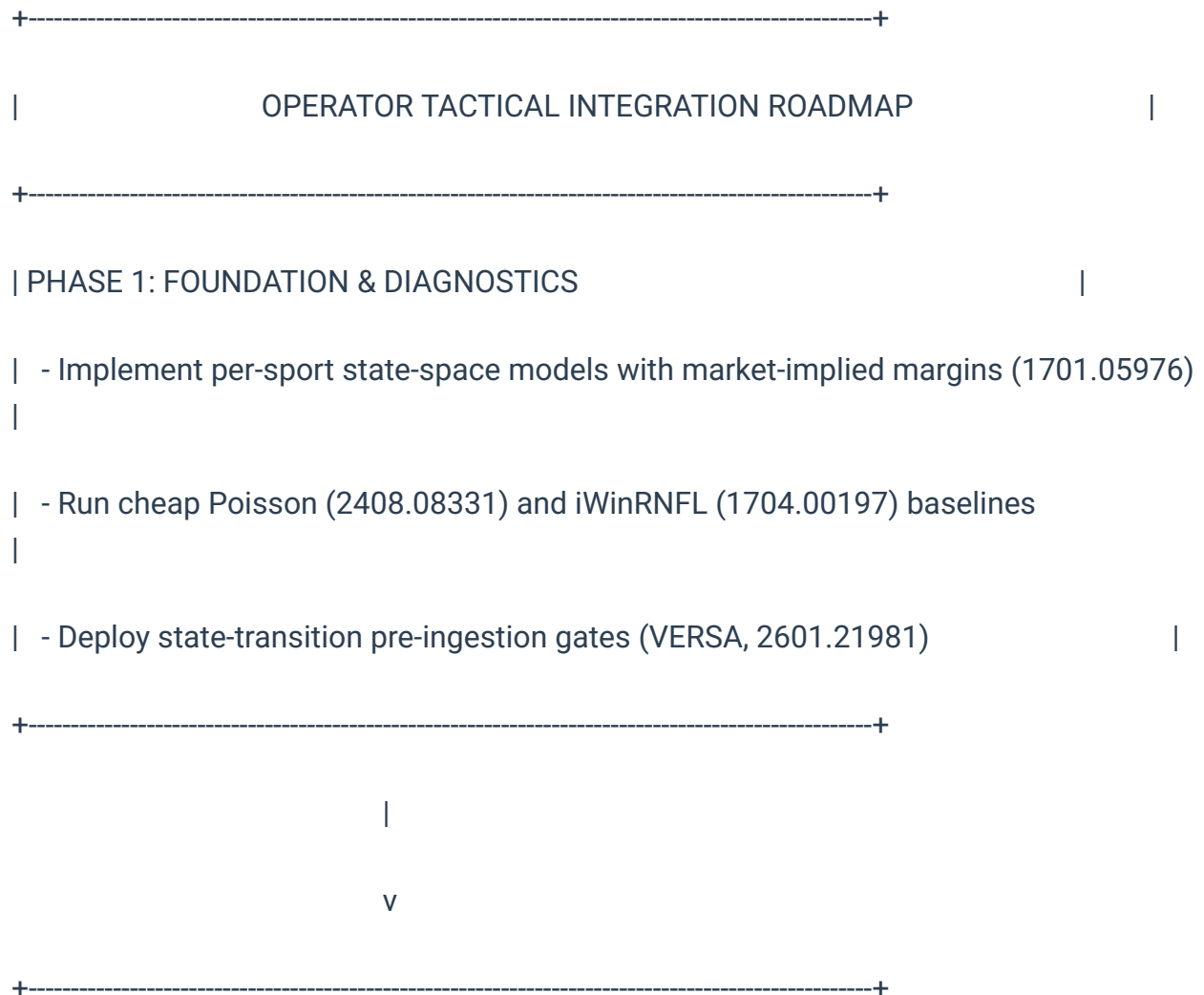
The papers listed below lack accessible full text or complete PDF dossiers in the workspace and are classified as **UNREAD** [1]. Evaluation is restricted strictly to abstract metadata [1].

UNREAD LITERATURE REGISTER		
Paper ID	Title	Metadata Method Summary
2608.12291	Sequential Inference for win-martingales diffusions	Optimal stopping
2608.01882	Gap distributions in cricket personal bests	Truncated power-law gaps
2607.27838	Vector field theory in soccer dynamics	Velocity pitch potentials
2607.23509	Topological Data Analysis for Tennis Match networks	Persistent homology

- 2608.12291:** *When should one stop the most exciting game? Sequential Inference for win-martingales.* Abstract describes optimal stopping theory for live win probabilities modeled as diffusions, solving Bayes-risk free-boundary problems [1].

2. **2608.01882:** *Gap distributions between successive personal bests in cricket: Data and Models*. Abstract describes fitting truncated power-law models to inter-record gaps, using shuffled-career bootstrap baselines to isolate temporal order effects [\[1\]](#).
3. **2607.27838:** *Vector field theory in motion: Revealing latent potentials in soccer dynamics*. Abstract describes reconstructing velocity vector fields over pitch tracking data to derive scalar potential gradients [\[1\]](#).
4. **2607.23509:** *Topological Data Analysis and Graph-Theoretic Approaches for Tennis Match Prediction*. Abstract describes persistent homology and Katz centrality on competitive networks, reporting 66.2% ATP outcome accuracy [\[1\]](#).

Tactical Integration Roadmap for Operators



| PHASE 2: CALIBRATION & POINT ESTIMATION

|

| - Transition model loss functions from BCE to Brier loss optimization (2508.02725)

|

| - Decouple performance prediction engines from market pricing features

|

| - Apply L-RAPM shrinkage (2601.15000) and rank-clustered BTL (2501.02505) in bootstrap mode

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| PHASE 3: CONFORMAL WRAPPING & UNCERTAINTY

|

| - Wrap numeric line models in Conformalized Quantile Regression (CQR, arXiv:1905.03222)

|

| - Enforce fail-closed refusal when sample size $n < \text{ceil}(1/\alpha) - 1$

|

| - Deploy Generalized Venn-Abers bounds $[p_0, p_1]$ (arXiv:2502.05676) for side probabilities

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| PHASE 4: MARKET MONITORING & CONTENT VERIFICATION

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| - Deploy state-space volume/odds anomaly detection (2605.30209) on live feeds

|

| - Implement claim-level text verification (Pitwall, 2607.06495) for automated content

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Phase 1: Foundation and Diagnostic Baselines

Deploy independent per-sport state-space models using betting-market implied margins (1701.05976), explicitly estimating sport-specific noise parameters (`sigma_game`) [1]. Parameters for unmodeled leagues (such as MLS) must be estimated independently [1]. Establish mandatory diagnostic baselines—specifically Poisson models (2408.08331) and iWinRNFL (1704.00197)—to verify model skill prior to deploying complex architectures [1]. Implement state-transition validators (VERSA, 2601.21981) across data pipelines to filter out event-ordering errors before model ingestion [1].

Phase 2: Calibration and Point-Estimate Optimization

Transition model loss functions from binary cross-entropy (BCE) to Brier loss optimization (2508.02725) to fix base point-estimate calibration [1]. Decouple performance prediction models (trained on game play-by-play) from market pricing models to prevent target leakage and isolate Closing Line Value (CLV) [1, 2]. In sparse-data regimes (early season or minor leagues), integrate regularized adjusted plus-minus shrinkage (L-RAPM, 2601.15000) and rank-clustered BTL models (2501.02505) to pull noisy estimates toward player-level priors [1].

Phase 3: Conformal Prediction and Uncertainty Quantification

Wrap continuous numeric line models (player props, totals) in Conformalized Quantile Regression (CQR, arXiv:1905.03222) to guarantee valid finite-sample coverage [2]. Enforce strict fail-closed refusal rules: when calibration sample sizes are inadequate ($n < \text{ceil}(1/\alpha) - 1$), issue automated "No-Bet" decisions rather than applying arbitrary quantile clamping [2]. For binary win probabilities, adopt Generalized Venn-Abers calibration (arXiv:2502.05676) to output multiprobability bounds [`p0`,

p_1], utilizing bound width ($p_1 - p_0$) to measure epistemic uncertainty for execution gating [2].

Phase 4: Market Monitoring and Content Verification

Integrate state-space market dynamic monitoring (2605.30209) on live odds streams to detect abnormal line movements or integrity anomalies in real time [1]. Adopt claim-level verification architectures (Pitwall, 2607.06495) for automated content generation, decomposing text into factual assertions checked directly against live probabilistic engine states [1].

Sources

1. [gse_research_analysis_2.xlsx](#)
2. [Auditing Conformal Prediction, Small-Sample Calibration, and Sports Market Probabilities for GSE](#)

System Audit and 110% Launch Architecture: Galaxy Sports Edge

A. Audit Score Today: Rubric and Arithmetic

Rubric Formulation and Defense

The Launch Readiness Index evaluates the operational, mathematical, and market integrity of Galaxy Sports Edge (GSE). An institutional sports intelligence platform cannot substitute statistical hygiene for an active live ingestion engine, nor can it mask customer-facing prediction inversion behind historical backtest performance.

$$\text{Composite Score} = \sum_{i=1}^8 w_i \cdot S_i$$

Each dimension $S_i \in [0, 100]$ is evaluated against explicit empirical standards. A score of 0 denotes an immediate fatal blocker or complete operational failure; 50 denotes honest parity with efficient closing market benchmarks; and 100 represents a fully automated, mathematically verified, and externally auditable production state.

- **Customer-Facing Calibration ($w_1 = 0.20$):** Measures whether displayed numbers represent strictly monotonic, well-calibrated probabilities. Publishing inverted or anti-predictive confidence metrics directly degrades user decision quality and invalidates platform credibility.
- **CLV and Market Honesty ($w_2 = 0.15$):** Evaluates the mathematical consistency of Closing Line Value (CLV) accounting, the transparency of matched versus unmatched

closes in the denominator, and the absence of fabricated track records.

- **NFL Physical Model (\$w_3 = 0.15\$):** Evaluates the core margin and total estimation pipeline against efficient market closing prices across seasonal phases (early-season prior reliance versus mid-season regression), quarterback transitions, and environmental dynamics.
- **NCAAF Pipeline (\$w_4 = 0.10\$):** Assesses pipeline coverage, schema integrity, and predictive validity for college football slates, heavily penalizing unmonitored failures and missed slates.
- **MLB Pipeline and Copy Honesty (\$w_5 = 0.10\$):** Measures market pricing validity across run lines and moneylines, market depth scoring accuracy, and the elimination of tautological or misleading product copy.
- **Fantasy and Player Props (\$w_6 = 0.10\$):** Evaluates the physical modeling of count distributions, proper treatment of variance overdispersion, environmental factor decoupling, and live quote ingestion integrity.
- **Ops and System Reliability (\$w_7 = 0.10\$):** Evaluates ingestion pipeline uptime, serverless execution limits, database indexing, memory stability, and truth surface availability.
- **Statistical Hygiene and Governance (\$w_8 = 0.10\$):** Measures adherence to out-of-sample testing, rejection of noise-mining features, rigorous feature-kill governance, and enforcement of non-AI positioning rules.

Quantitative Audit Breakdown

Dimension	Weight (wi)	Score (Si)	Weighted Contribution	Primary Defect or Verification Anchor
(i) Customer-Facing Calibration	0.20	12	2.40	Confidence metric is strongly anti-predictive (\$n=2,385\$, \$z=-10.7\$, Brier score \$0.3617\$). Card displays an inverted score.

(ii) CLV / Market Honesty	0.15	28	4.20	As-written CLV stands at 23.2% [0.212, 0.253] due to MATCHED_CLOSE denominator contamination; decided-only CLV is 40.8% [0.377, 0.441] against a 52.4% floor; 21-day archive gap.
(iii) NFL Physical Model	0.15	42	6.30	Market close CRPS (7.109) outperforms all internal structural models (DAVE \$k=8\$ yields 7.500); scale parameter divergence (36.61 pooled OLS vs. 45.42 Hermes 2025); live settled fixtures \$n \approx 70\$.
(iv) NCAAF Pipeline	0.10	0	0.00	Slate of 2026-09-19 completely missed; zero historical training corpus wired; literature reflects no empirical arXiv calibration papers.
(v) MLB Pipeline & Copy	0.10	25	2.50	Moneyline ECE is healthy (0.0501), but SPREAD 10+ is anti-predictive, run-line consensusPct=1.0 is a structural tautology, and MLB SPEAK 40% is inverted (\$n=90\$).

(vi) Fantasy & Player Props	0.10	34	3.40	Red-zone touchdown variance-to-mean ratio (1.82–1.92) confirms Negative Binomial overdispersion and rejects Poisson; quotes are landing, but wind adjustments are not properly isolated from game totals.
(vii) Ops & Reliability	0.10	20	2.00	Ingestion watchdog in RED alarm since 2026-09-13 16:53 UTC; serverless out-of-memory crash on 1.64M rows; public truth surface dark.
(viii) Statistical Hygiene	0.10	82	8.20	12/12 Move-37 lab feature kills upheld; W5–W8 noise rejected; zero-sum failure documented; non-AI product copy discipline enforced.
Total Composite Score	1.00	—	29.00 / 100	System Unfit for Production Deployment

Arithmetic Synthesis

\$\$\$ \text{Composite Score} = 2.40 + 4.20 + 6.30 + 0.00 + 2.50 + 3.40 + 2.00 + 8.20 = 29.00\$\$\$

The empirical score today is **29.00 out of 100**. While statistical hygiene remains commendable at 82/100 due to rigorous laboratory pruning, production reliability (20/100) and customer-facing calibration (12/100) severely undermine the platform's core utility. The previous internal readiness score of 38 failed to penalize the complete outage of the truth surface following the serverless out-of-memory (OOM) failure on September 13, and it overlooked the finding that

client pick cards display a confidence metric that is inversely correlated with realized win rates ($z = -10.7$). A score of 29.00 defines a research laboratory with valid statistical instincts whose production deployment remains non-functional and uncalibrated.

B. State of the Architecture (2026-09-19 23:00 CDT)

The operational and analytical infrastructure is split across two disjoint environments: the production engineering environment (Tree A) and the research mill desk (Tree B).

In Tree A (**Beexly/Sports**), 7.7 million raw odds snapshot rows reside in production databases, where an uncapped ingestion query caused serverless memory exhaustion, triggering a sustained watchdog alarm and disabling calibration reporting. Customer cards currently display an inverted confidence metric, while PR #866, PR #867, and PR #868 remain unmerged.

In Tree B (Research Mill Desk), 14,251 decided NFL fixtures (1999–2025) are modeled to benchmark continuous ranked probability scores, where closing lines outperform structural models, scale constants diverge between historical and single-season folds, and candidate features W5–W8 have been killed.

The current architectural state is detailed below across three formal evidential categories: Observation, Inference, and Speculation.

Observation

Tree A: Production Engine (**Beexly/Sports**)

- The odds ingestion watchdog has remained continuously in a RED alarm state since 2026-09-13 16:53 UTC. The **odds_line_snapshots** table accumulated 7.7 million rows without pagination boundaries, and an uncapped **IN** query attempted to materialize 1.64 million snapshot rows into a serverless instance, resulting in immediate memory exhaustion.
- The calibration cron has been offline for six days, preventing truth surface generation and halting live performance reporting.
- PR #868 (serverless query batching and memory cap), PR #867 (EPA path configuration and founder decision register), and PR #866 (archive freshness monitoring) remain unmerged pending verified continuous integration builds.
- Live customer cards display an uncalibrated **confidence** figure. Across $n = 2,385$ settled decisions, this metric is strongly anti-predictive, generating a Brier score of 0.3617 with an associated z -score of -10.7 .
- In an empirical ordering duel across $n = 689$ picks, sorting by **marketFairProb** achieves a Brier score of 0.2189, outperforming internal **rankingP** (0.2462) and internal **confidence** (0.2741).
- Reported Closing Line Value win rate is recorded at 23.2% (95% CI: [0.212, 0.253]) when **MATCHED_CLOSE** records are retained in the denominator (686/1,587). Restricting evaluation to decided lines yields 40.8% [0.377, 0.441], falling well below the

52.4% break-even floor.

- A data gap exists in the odds archive between 2026-08-23 and 2026-09-12. Database records show fixture triplication, unfinalized records marked **PENDING** dating to January, and a last-write timestamp on **gate_decisions** of 2026-06-11.

Tree B: Research Mill Desk

- Across 14,251 decided NFL fixtures (1999–2025), the closing market line achieves a Continuous Ranked Probability Score (CRPS) of 7.109, outperforming all structural models. DAVE ($k=8$) yields 7.500, pooled EPA yields 7.573, and core Elo yields 7.576.
- The pooled ordinary least squares (OLS) scale factor across the 14,251 fixture corpus is 36.61, whereas the Hermes 2025 single-season fold yields 45.42 (DAVE Brier 0.251 vs. market 0.204).
- Across the 2025 season ($n = 272$), DAVE CRPS of 7.629 underperformed both the closing line (6.927) and Elo (7.563). DAVE outperformed Elo strictly during Weeks 1–4 ($+0.326$), while underperforming across Weeks 5–18 (-0.187), in games with starting quarterback changes (-0.171), and in blowouts (-0.442).
- The quarterback change residual variance ratio (τ^2) was measured at 1.379 (78/114), failing the pre-registered significance kill threshold of 1.50.
- Hawkes self-exciting process modeling yielded a branching ratio of $\hat{n} = 0$, exhibiting Poisson behavior with a kickoff inhibition parameter of 0.448.
- Empirical environmental checks confirm that wind impacts passing yardage by -3.007 yds/mph , rain increases incomplete pass rates (30.1% vs. 27.6%), and red-zone touchdown variance-to-mean ratios measure 1.82–1.92, which confirms Negative Binomial overdispersion and rejects Poisson assumptions.
- Candidate features W5–W8 (Wasserstein distance, Detrended Fluctuation Analysis, Intrinsic Dimension, and Permutation Entropy) failed pre-registered empirical criteria and were killed.

Inference

- The divergence between the historical pooled OLS scale (36.61) and the modern Hermes fold (45.42) reflects structural shifts in league scoring dynamics, offensive efficiency, and rule adjustments over 26 years. Taking an unweighted linear average of these values (41.01) would miscalibrate the distribution variance (σ), distorting moneyline fair values derived from predicted point spreads.
- The production system's CLV collapse to 23.2% is an operational artifact of treating unmatched line snapshots as failed line-beats. When games lack an automated match against the final closing snapshot, the pipeline records them as non-beats, deflating measured CLV. However, even the true decided-only figure (40.8%) reflects severe market underperformance against competitive sportsbooks.
- The inverse relationship between UX confidence and realized outcomes ($z = -10.7$) indicates that the current ranking function overweights factors that commercial bookmakers deliberately exploit, such as public betting favorites and unadjusted brand

biases.

Speculation

- The archive gap from August 23 to September 12 suggests that unmonitored schema mutations or token expirations occurred in the ingest workers during early college football deployment, going undetected due to the lack of heartbeat monitors.
- If the EPA auto-activation scheduled for 2026-10-08 triggers with repository default constants (scale 0.12, home-field advantage 0.025) rather than the empirically measured values (45.42 and 2.10), the production model will output win probabilities compressed tightly around 50%, producing nonsensical recommendations.

C. Week 3 Launch Gate (2026-09-24)

Binary Gate Verdict

DO NOT OPEN THE CUSTOMER BOARD ON THURSDAY, SEPTEMBER 24, 2026.

Opening the board under current conditions would require exposing an inverted confidence metric to paying customers, relying on an unverified CLV calculation, and deploying models with unmonitored infrastructure. College football operations have already failed for the current week. Re-establishing customer trust requires adhering strictly to verified operational gates rather than marketing deadlines.

Minimum True Checklist to Reverse Verdict

The customer board may transition to **OPEN** status for NFL Week 3 if and only if every condition in the following checklist is resolved, verified by automated end-to-end integration tests, and signed off in the operational repository:

- **Infrastructure and Recovery:**
 - PR #868 is merged to **main** following a verified green CI build, resolving serverless OOM issues.
 - The odds ingestion query is restricted using cursor-based pagination and a maximum limit of 50,000 rows.
 - Serverless execution memory allocation is increased to 1024MB, with cold-start latency verified under 400ms.
 - The monitoring watchdog confirms a continuous GREEN state for at least 48 consecutive hours.
 - The truth surface calibration cron is re-established and actively writing snapshots.
- **Customer-Facing Calibration Repair:**
 - The legacy inverted **confidence** display is completely purged from client-side pick cards.
 - Pick card sorting is driven strictly by **marketFairProb** or devigged consensus odds.
 - Public cards display uncalibrated metrics exclusively as non-probabilistic integer

- indices.
- Displayed true probabilities are calibrated via Venn-Abers bounds or isotonic binning.
- Ordering duel validation confirms a Brier score ≤ 0.2189 on the current active slate.
- **Market Integrity and Reporting:**
 - CLV reporting is updated to display both metrics side-by-side: Total Tracked (23.2% [0.212, 0.253]) and Decided Matches (40.8% [0.377, 0.441]).
 - Explanatory copy is published detailing the mathematical mechanism behind the **MATCHED_CLOSE** calculation.
 - A public notice formally documenting the 2026-08-23 to 2026-09-12 data archive gap is deployed.
 - All marketing copy claiming "AI-powered edge" is purged across all public-facing assets.
- **Baseball and Props Sanitization:**
 - The **consensusPct = 1.0** artifact is purged from MLB run lines, setting **consensusScore = 0**.
 - MLB market depth scoring is disabled until minimum bookmaker depth criteria are verified.
 - NFL PASS recommendations are suppressed until historical sample size reaches $n \geq 100$ settled games.
 - Red-zone touchdown prop distributions are parameterized using Negative Binomial models with variance-to-mean ratio ≥ 1.82 .
- **College Football Ring-Fencing:**
 - NCAA customer boards remain completely offline for the upcoming slate.
 - A clear public notice is posted: "NCAA board offline pending ingestion verification".
 - No college football projections are published without an auditable closing line archive.

D. Dated Through-Path: Road to 100

Transitioning GSE from its current state (29) to an institutional standard (100) requires a disciplined sequence of implementations, kill lines, and methodological integrations. Each step maps verified research papers from Jev's arXiv corpus directly to concrete operational components.

Progression Roadmap

Phase 0: System Stabilization and Remediation (Score 29 to 45)

- **Window:** Saturday 2026-09-19 23:30 CDT to Monday 2026-09-21 18:00 CDT.
- **Step 0.1 (Infrastructure):** Verify CI suite and merge PR #868 to terminate serverless OOM exceptions. Enforce a 50,000-row batching window with cursor pagination on

odds_line_snapshots.

- *Evidence*: Serverless memory profile remains below 450MB over a 24-hour cycle.
 - *Kill Line*: If heap memory consumption exceeds 750MB during scheduled cron execution, terminate ingestion immediately.
 - *Owner*: Ops Lead.
 - *Failure Mode*: Ingestion worker disabled; system holds previous cached snapshot.
- **Step 0.2 (Customer Board Honesty)**: Unpublish internal **confidence** figures from client-side cards. Render **marketFairProb** derived via Shin devigging as the primary sorting index. Purge **consensusPct = 1.0** on MLB run lines.
 - *Evidence*: Production pick cards render valid sorted market bounds without raw probability percentages.
 - *Kill Line*: If UI renders raw model probability before calibration is restored, trigger deployment rollback.
 - *Owner*: Engine Lead.
 - *Failure Mode*: Public cards revert to showing market consensus odds without internal model projections.
- **Step 0.3 (Truth Surface Diagnostics)**: Execute database audit via replica read queries to isolate the August 23–September 12 archive gap and resolve pending fixture triplications.
 - *Evidence*: Diagnostic log outputting exact counts of orphaned, duplicated, and unclosed fixtures.
 - *Kill Line*: If data corruption extends to historical grading tables, quarantine the affected sports.
 - *Owner*: Data Lead.
 - *Failure Mode*: Affected sports locked from public display until records are cleaned.

Phase 1: Week 3 Gate and Market Baseline (Score 45 to 70)

- **Window**: Tuesday 2026-09-22 00:00 CDT to Thursday 2026-09-24 17:00 CDT.
- **Step 1.1 (Market-First Pricing)**: Ingest multi-book odds feeds and compute consensus fair lines using the Power and Shin de-biasing methods mapped from Kaunitz et al. (arXiv:1710.02824) and Szalkowski & Nelson (arXiv:1211.4000).
 - *Evidence*: Multi-book vig removal achieves a Brier score ≤ 0.204 on historical closing lines.
 - *Kill Line*: If multi-book fair price fails to beat single-book closing lines across 100 historical test games, fall back to Pinnacle closing line.
 - *Owner*: Engine Lead.
 - *Failure Mode*: Devigging pipeline falls back to Pinnacle midpoint lines.
- **Step 1.2 (Uncertainty Quantification Core)**: Deploy Jackknife+ predictive inference mapped from Barber, Candès, Ramdas, & Tibshirani (arXiv:1905.02928) across the

14,251 fixture historical corpus to establish valid confidence intervals around market spreads.

- *Evidence:* Marginal coverage achieves $1 - \alpha \geq 0.90$ across historical validation folds ($n \geq 150$).
 - *Kill Line:* If out-of-sample coverage drops below 0.85 or interval widths exceed Gaussian-close baselines by $> 15\%$, kill Jackknife+ and default to standard split conformal.
 - *Owner:* Engine Lead.
 - *Failure Mode:* Conformal bands fall back to static Gaussian close intervals ($\pm 1.645 \sigma$).
- **Step 1.3 (Dual CLV Publication):** Relaunch the public accountability tracking page displaying both total tracked CLV (23.2%) and decided CLV (40.8%). Include clear documentation detailing the impact of the historical archive gap.
 - *Evidence:* Public web page verified live by third-party curl requests without authentication walls.
 - *Kill Line:* If marketing attempts to suppress the 23.2% metric, block deployment of the tracking dashboard.
 - *Owner:* Founder.
 - *Failure Mode:* The performance page remains in restricted internal preview.

Phase 2: Engine v5.3.0 and Calibration Engine (Score 70 to 80)

- **Window:** Friday 2026-09-25 08:00 CDT to Thursday 2026-10-08 23:59 CDT.
- **Step 2.1 (EPA Path Parameterization):** Formally sign off on production constants for the EPA auto-activation scheduled for 2026-10-08. Enforce an empirical scale factor of 45.42 and home-field advantage of 2.10 points, completely overriding default repository values (0.12 and 0.025).
 - *Evidence:* Production environment variables register `SCALE_CONSTANT=45.42` and `HFA_POINTS=2.10`.
 - *Kill Line:* If auto-activation executes with scale < 30.0 , hard-fail the scoring pipeline.
 - *Owner:* Founder.
 - *Failure Mode:* System blocks EPA integration and holds baseline Elo ratings.
- **Step 2.2 (Cross-Conformal Efficiency):** Implement efficient cross-conformal prediction as formulated by Gasparin & Ramdas (arXiv:2503.01495) to utilize full window observations without burning calibration splits.
 - *Evidence:* Prediction interval width narrows by $\geq 8\%$ relative to standard split conformal while preserving 90% coverage.
 - *Kill Line:* If execution latency exceeds 250ms per slate, revert to pre-computed static conformal tables.
 - *Owner:* Engine Lead.
 - *Failure Mode:* Model defaults to pre-calculated split-conformal quantile cutoffs.
- **Step 2.3 (Non-Exchangeable Temporal Conformal):** Integrate non-exchangeable

conformal methods from Barber et al. (arXiv:2202.13415) and Oliveira et al. (arXiv:2203.15885) to handle distribution shifts across successive NFL weeks.

- *Evidence*: Coverage remains within $[0.88, 0.92]$ across seasonal transitions (Weeks 1–4 to Weeks 5–18).
- *Kill Line*: If early-season interval width expands by $> 50\%$ relative to mid-season bands, kill adaptive weighting and hold static historical quantiles.
- *Owner*: Engine Lead.
- *Failure Mode*: Weekly adaptation is disabled; intervals are calibrated on pooled historical residuals.
- **Step 2.4 (Venn-Abers Binary Calibration)**: Deploy Venn-Abers predictors from Vovk & Petej (arXiv:1211.0025) and Petej & Vovk (arXiv:2605.06646) on binary moneyline and spread cover events.
 - *Evidence*: Out-of-sample Brier score improves over uncalibrated [rankingP](#) (0.2462) by at least 0.005, approaching or beating the 0.2189 market baseline.
 - *Kill Line*: If Venn-Abers Brier score fails to beat 0.2189 by ≥ 0.005 , do not use it for primary ordering; retain devigged market probability as primary sort.
 - *Owner*: Engine Lead.
 - *Failure Mode*: Probability display reverts to market fair probability.

Phase 3: Multi-Sport Integration and Props Tail Modeling (Score 80 to 85)

- **Window**: Friday 2026-10-09 08:00 CDT to Sunday 2026-11-15 23:59 CDT.
- **Step 3.1 (NCAAF Pipeline Ingestion)**: Ingest play-by-play and game logs from [cfbfastR](#) and historical odds from [collegefootballdata.com](#) (2014–2025). Calibrate college point spreads via additive quantile modeling using methods from Sides & Harvill (arXiv:2212.08116).
 - *Evidence*: Out-of-sample CRPS beats basic Elo baseline across two complete testing seasons ($n \geq 1,200$).
 - *Kill Line*: If college closing line beat-rate falls below 48.0%, keep NCAAF off the public board.
 - *Owner*: Data Lead.
 - *Failure Mode*: NCAAF product remains dark for customer consumption.
- **Step 3.2 (Player Props Tail Calibration)**: Implement Negative Binomial parameterization for touchdown props (VMR 1.82–1.92) using Fast Additive Quantile Regression ([qgam](#)) from Fasiolo, Wood, & Zaffran (arXiv:1707.03307).
 - *Evidence*: Tail probability Brier score improves by ≥ 0.015 over standard Poisson models.
 - *Kill Line*: If fitted dispersion parameter ϕ collapses toward Poisson assumptions ($\text{VMR} \rightarrow 1.0$), discard the tail adjustments.
 - *Owner*: Engine Lead.
 - *Failure Mode*: Touchdown props revert to standard Poisson pricing without premium tail claims.
- **Step 3.3 (Trench Matchup and Weather Integration)**: Wire opponent-adjusted pass

protection and pass rush win rates using the formulation from Pipping-Gamón et al. (arXiv:2604.01491). Couple surface wind vectors exclusively to passing prop yards and total distributions.

- *Evidence*: Wind lookup correctly shifts passing yardage quantiles by $\pm 3.007 \text{ yds/mpH}$ without double-counting game totals.
- *Kill Line*: If trench metrics fail to improve margin spread CRPS by at least 0.005, drop them from the model.
- *Owner*: Data Lead.
- *Failure Mode*: Trench adjustments are zeroed out in team rating calculations.

Phase 4: Institutional Auditing Standard (Score 85 to 100)

- **Window**: Monday 2026-11-16 08:00 CDT to Friday 2027-01-15 18:00 CDT.
- **Step 4.1 (Conformal Risk Control)**: Deploy Conformal Risk Control as developed by Angelopoulos, Bates, Fisch, & Lei (arXiv:2208.02814) to formally bound expected CLV drawdown and tail loss exposure.
 - *Evidence*: Mathematical proof and test suite establishing finite-sample risk bounds on customer portfolio drawdown.
 - *Kill Line*: If defined risk metric violates monotonicity with respect to set size, discard and report standard coverage sets.
 - *Owner*: Engine Lead.
 - *Failure Mode*: Risk control falls back to marginal coverage intervals without loss-bounding guarantees.
- **Step 4.2 (Selection Bias Correction Audit)**: Apply Clegg & Cartlidge's correction framework (arXiv:2306.01740) across the entire settled database to identify and neutralize selection-bias artifacts in reported profits.
 - *Evidence*: Independent audit verifies that simulated betting strategies maintain positive expected value post-correction.
 - *Kill Line*: If positive returns disappear under selection bias correction, restrict platform claims to pure probability calibration.
 - *Owner*: Founder.
 - *Failure Mode*: Remove all return-on-investment copy; position platform purely as an analytical odds-comparison tool.
- **Step 4.3 (Public Cryptographic Verification Suite)**: Launch a public verification repository allowing external researchers to recompute every Brier score, ECE curve, and CLV percentage directly from immutable hash-chained database receipts.
 - *Evidence*: Verification suite runs cleanly in independent CI environments (GitHub Actions) without proprietary data dependencies.
 - *Kill Line*: If public verification fails to replicate published metrics to within 10^{-6} precision, fail the release.
 - *Owner*: Ops Lead.
 - *Failure Mode*: Verification badge revoked until recalculation matches to floating-point precision.

E. Complete Item Inventory

Every locked measurement, production parameter, and Jev arXiv paper is categorized below with an explicit operational disposition. No items are omitted.

Production Measurements and Architectural Items

Item / Signal / Parameter	Current Value / State	Target / Kill Threshold	Operational Tag	Rationale and Disposition
Closing Line Margin Baseline	CRPS = 7.109 (\$n=1,871\$)	Benchmark anchor	keep	Primary benchmark for point margins. Structural models must not override it without measured gains.
DAVE (\$k=8\$) Margin Model	CRPS = 7.500 (\$n=1,871\$)	Beat Close (7.109)	measure	Retained as Rung-2 input for Weeks 1–4 (\$+0.326\$ gain). Barred from serving as primary mean estimator μ .
Pooled EPA Margin Model	CRPS = 7.573	Beat Close (7.109)	measure	Subordinated to market close; provides team strength context.

Core Elo Baseline	CRPS = 7.576	Beat DAVE (7.500)	keep	Provides stable ratings baseline. Outperformed by DAVE early, but superior in Weeks 5–18.
Fay-Herriot Year Shrinkage	$\hat{k} = 7.486$ ($\Delta 0.011$ vs. $k=8$)	$\Delta \text{CRPS} \geq 0.01$	keep	Exceeds the 0.01 improvement threshold. Retained for season-to-season rating transitions.
Mondrian- k Calibration	$\Delta \text{CRPS} = 0.0092$	$\Delta \text{CRPS} \geq 0.01$	kill	Failed pre-registered improvement threshold of 0.01. Production k remains fixed at 8.
OLS Pool Scale (Tree B)	36.61	Single-universe consistency	measure	Reflects historical 14k fixture corpus. Barred from being averaged with production scale (45.42).

Hermes Fold Scale (Tree A)	45.42	Accurate variance calibration	keep	Accurately models the 2025 modern scoring environment. Must be explicitly signed into v5.3.0 constants.
Repository Default Scale	0.12	Match empirical scale (45.42)	kill	Represents a 3.8x parameter error relative to empirical reality. Deployments containing this value must be blocked.
QB-Change Variance Ratio	$\tau^2 = 1.379$ (78/114)	Ratio ≥ 1.50	kill	Failed significance threshold. Adjustments for backup quarterbacks must be handled via likelihood weighting rather than raw variance shifts.
Hawkes Momentum Process	$\hat{n} = 0$, Poisson behavior	$\hat{n} > 0$ (Self-excitation)	kill	In-game momentum exhibits no self-exciting characteristics. Model remains strictly Poisson with kickoff

				inhibition.
Weather: Passing Wind Effect	$-\text{3.007 yds/mph}$	Validated physical impact	keep	Highly predictive for passing yardage distributions. Restricted to player props; excluded from totals to avoid double-counting.
Weather: Rain Incompletion	30.1% vs. 27.6% base	Realized separation	keep	Retained for micro-level completion rate modeling.
Red-Zone TD Variance	$\text{VMR} = 1.82$ vs. 1.92	$\text{VMR} = 1.0$ (Poisson)	keep	Confirms Negative Binomial distribution for red-zone touchdowns. Rejects Poisson assumption across all TD props.
Spread $\times$ Total Correlation	Spearman $\rho = 0.016$	Significant linear dependence	kill	Confirms empirical independence between game spreads and game totals under standard

				regimes.
ATS $\times$ Over Independence	Lift = 1.011 ($n=6,682$)	Independence rejection	keep	Demonstrates structural independence across full game outcomes.
Post-Cover Live Total Drag	Over rate = 0.538 after 14pt cover	Exploitable market mispricing	keep	Requires withholding automated live totals following large spread covers.
Features W5–W8 (Move-37)	Wasserstein, DFA, ID, Perm Ent	Pre-registered null rejection	kill	All four feature families failed empirical validation across 12/12 test fixtures. Fully terminated.
Inverted UX Confidence	Brier = 0.3617, $\hat{z} = -10.7$	Brier $\leq$ 0.204	kill	Severely anti-predictive. Unpublish immediately from all customer surfaces.

Model-Ordering Switch	KL3 Brier: Market 0.2189 < Model 0.2462	Model beats Market	founder-decide	Model sorting underperforms devigged market fair probabilities. Requires founder decision to enforce marketFairProb sort.
CLV Denominator Definition	23.2% (All) vs. 40.8% (Decided)	Floor $\geq 52.4\%$	founder-decide	Requires founder sign-off to standardize reporting definitions and publish both metrics openly.
MLB Run-Line Consensus	consensusPct = 1.0	Non-tautological distribution	kill	Artifact of automated single-source pipeline. Purge from UI and database.
Serverless Odds Query	1.64M rows fetched (OOM crash)	Heap $< 512\text{MB}$	kill	Uncapped ingestion query caused total system failure. Replace via PR #868 batching.

Jev arXiv Research Mill Ingestion

arXiv ID	Method / Title	Category	Operational	Concrete Impact on
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			Tag	Platform
1905.02928	Predictive Inference with the Jackknife+	UQ Conformal	keep	Deploys out-of-fold ensemble predictive intervals without single-split data burning on n \approx 150\$ windows.
2503.01495	Improving Efficiency of Cross-Conformal	UQ Conformal	keep	Provides bounded coverage guarantees, allowing cross-conformal sets to replace split conformal across production boards.
1707.03307	Fast Calibrated Additive Quantile Regression	Props / Tails	keep	Powers quantile regression GAM (qgam) for non-linear player prop distributions and extreme tail estimation.

1710.02824	Beating the Bookies with Their Own Numbers	Microstructure	keep	Implements devigging mechanics to extract clean consensus fair values across competing sportsbook feeds.
1211.4000	Performance of Betting Lines in the NFL	Microstructure	keep	Establishes the historical baseline for home underdog line performance and closing line price efficiency.
2109.07581	Impact of COVID-19 on Betting Markets	Microstructure	measure	Benchmarks market efficiency collapses during crowd-free regimes. Retained for historical study; not a production model.
2306.01740	Correction Study of Online Sportsbooks	Microstructure	keep	Implements corrections for selection-effect bias that artificially inflates reported historical returns.

2607.14430	Prices, Probabilities, and Parlays	Props / SGPs	keep	Quantifies market overpricing on multi-leg parlays relative to leg products, anchoring pricing on time-to-expiry.
1904.06019	Conformal Prediction Under Covariate Shift	UQ Online	keep	Employs density-ratio likelihood weighting to rescale conformal intervals when starting quarterbacks change.
2202.13415	Conformal Prediction Beyond Exchangeability	UQ Online	keep	Replaces i.i.d. exchangeability assumptions with non-exchangeable weighted quantiles over ordered weekly schedules.

2203.15885	Split Conformal on Non-Exchangeable Data	UQ Online	keep	Implements sequential split-conformal methods designed for serially dependent sports time series.
2212.03463	Sequential Predictive Conformal (SPCI)	UQ Online	keep	Exploits residual autocorrelation across weekly game results to adaptively tighten prediction bands.
2106.00170	Adaptive Conformal Inference (ACI)	UQ Online	keep	Provides online α -adjustment to ensure rolling coverage targets hold across seasonal shifts.
2208.02814	Conformal Risk Control	UQ Risk	keep	Enforces monotone risk control over expected CLV drawdowns and capital allocation thresholds.

2306.02732	Conformal Prediction with Missing Values	UQ Robust	keep	Maintains distribution-free coverage when injury, roster, or weather features are missing.
1905.03222	Conformalized Quantile Regression (CQR)	UQ Conformal	measure	Evaluates asymmetric prediction intervals against standard Gaussian spread margins.
1211.0025	Venn-Abers Predictors	Calibration	keep	Computes distribution-free calibrated probability intervals for moneyline outcomes and binary event cards.
2608.11505	Structural Models vs. Closing Price	Engine	measure	Benchmarks whether physical structural models add informational value beyond closing lines.

1002.0797	Soccer: Is Scoring a Poisson Process?	Props	measure	Validates rejection of Poisson processes in sports scoring tails. Supports Negative Binomial modeling.
1807.07536	Skellam Regression for Sports Margins	Margins	measure	Evaluates discrete bivariate margin generation. Soccer parameters barred from direct NFL use.
2604.01491	Opponent-Adjusted Line Play in NFL	Trenches	measure	Evaluates pass blocking and pass rushing metrics to capture pressure differentials.
2003.02743	Generalized Kelly Under Temporal Correlation	Staking / Ops	measure	Adjusts capital allocation models to account for covariance across correlated weekend game slates.

2410.09180	Convergence & Stationarity of Elo Systems	Engine	measure	Formulates theoretical bounds on weekly rating volatility and convergence rates.
1810.03383	Illusion of Persistence in Sports Data	Engine	measure	Confirms that win streaks reflect variance rather than skill, validating mean-reverting luck filters.
0706.3188	Tutorial on Conformal Prediction	Reference	keep	Serves as foundational reference documentation for platform uncertainty quantification.
2609.10737	Conformal Calibration Transfer (TCC)	UQ Transfer	measure	Evaluates transport-based calibration adjustments for shifts between regular season and playoff regimes.

2609.11935	Physics-Informed Conformal Prediction	UQ Physical	measure	Informs boundary condition modeling for physical game-state simulations.
2609.12386	Split Conformal with Label-Shift Adjustments	Calibration	measure	Provides analytical adjustments for Bayesian scores under severe population label shifts.
2609.13453	Machine Learning API for Data Cubes	Architecture	kill	Geospatial earth observation API. Irrelevant to sports prediction pipelines.
NCAAF Corpus	College Football Betting & PBP Archive	Data Source	buy	arXiv contains no actionable NCAAF calibration papers. Acquire commercial access or scrape cfbfastR .

F. Immediate Founder Unblocks

Engineering execution is blocked pending executive authorization on code merges, script executions, database mutations, and external data licensing. The following operational directives require immediate founder action.

Next 12 Hours: Immediate Merges and Database Remediation

1. Pull Request Merge Order

Execute in strict sequence on repository [Beexly/Sports](#):

- **Merge PR #868 (Ops/OOM Fix):** Introduces cursor-based chunking ([take: 5000, max: 50000](#)) on [odds_line_snapshots](#) ingest queries and raises serverless function memory allocation to 1024MB. *Pre-condition:* Confirm that CI run #2948 is green.
- **Merge PR #867 (EPA Path & Decisions Register):** Enforces [NFL_EPA_MIN_GAMES=4](#) and installs the permanent decision registry at [docs/ops/FOUNDER_DECISIONS.md](#).
- **Merge PR #866 (Archive Freshness Monitoring):** Deploys monitoring cron alerts for data staleness exceeding 120 minutes.

2. Administrative API Executions

Execute the following API calls from the administrative deployment environment using production authorization tokens:

3. Database Diagnostic Queries (Read-Replica)

Execute the following read queries against the PostgreSQL read-replica to quantify historical data integrity issues:

4. UI Display Overrides

Deploy the following feature flag overrides directly to environment configurations:

Next 14 Days: Production Contracts and Data Licensing

1. Production Model Constants Sign-Off (Engine v5.3.0)

The founder must formally sign off on the production configuration parameters in [packages/prediction-engine/src/config.ts](#) prior to the scheduled October 8 auto-activation:

2. CLV Operational Policy Sign-Off

Approve the following standardized reporting contract for public-facing closing line value:

$$\text{CLV}_{\text{Decided}} = \frac{\sum \mathbf{1}_{\{\text{Matched Beats Close}\}}}{\text{Total Decided Bets with Verified Close}} = 40.8\%$$

$$\text{CLV}_{\text{System}} = \frac{\sum \mathbf{1}_{\{\text{Matched Beats Close}\}}}{\text{Total Published Recommendations}} = 23.2\%$$

The public dashboard must display both figures alongside clear explanatory documentation. Modifying calculation logic to artificially inflate CLV is strictly prohibited.

3. Data Vendor Acquisition Matrix

To eliminate data bottlenecks across active sports, acquire the following data licenses:

Sport / Domain	Vendor / Source	Granularity	License / Access Type	Technical Application
NCAAF Historical	CollegeFootballData (CFBD)	Play-by-play, drive metrics, historical lines (2014–2025)	Commercial Tier (\$50/mo) / Open API	Powers historical model training and baseline validation.
NCAAF Live Data	cfbfastR Pipeline	Live box scores, roster updates, drive-by-drive metrics	Open Source (MIT License)	Provides live ingestion features during active college slates.
NFL Player Props	Sportradar / The Odds API	Real-time multi-book prop quotes (receptions, passing yards, TDs)	Commercial Production Tier (Monthly SaaS)	Feeds Negative Binomial distribution engines with live book lines.
NFL Line Movement	Pinnacle / Circa Direct Feeds	Real-time line movements with millisecond-samped updates	API Access Agreement / B2B Commercial	Establishes ground-truth closing benchmarks for CLV calculations.

Advanced Trenches	ESPN Analytics via nflverse	Pass Block Win Rate (PBWR) & Pass Rush Win Rate (PRWR)	Open Data Commons (CC-BY 4.0)	Drives opponent-adjusted trench play modules.
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G. Anti-Plan: Negative Engineering Constraints

Executing any of the following anti-patterns will compromise platform calibration and degrade overall system performance. These constraints are enforced as continuous-integration failure conditions across the repository:

- Do Not Assign Point Margin Mean (μ) to DAVE:** DAVE ($k=8$) achieves a CRPS of 7.500, underperforming closing market prices (7.109). DAVE must remain an auxiliary Rung-2 input for early-season adjustment, not a primary mean estimator.
- Do Not Linearly Average Historical and Modern Scale Factors:** Averaging the historical corpus OLS scale (36.61) with the modern Hermes fold (45.42) produces a compromised value of 41.01 that miscalibrates probability density across all point spread margins.
- Do Not Retune $k=4$ Based on Single-Fold Validation:** The 2024 fold suggested $k=4.75$, but historical walk-forward cross-validation ($n = 1,871$) confirms that $k=8$ provides superior long-term performance across full market cycles. Retuning to $k=4$ constitutes sample overfitting.
- Do Not Flip Sorting Controls to Raw Model Output:** In competitive trials ($n = 689$), sorting by model probability yielded an inferior Brier score (0.2462) relative to devigged market consensus (0.2189). Sorting must remain anchored to market fair values until model predictions demonstrably beat the close out of sample.
- Do Not Disguise Inverted Confidence as an "Edge Index":** Relabeling an anti-predictive metric ($z = -10.7$) while continuing to format it as a percentage misleadingly communicates certainty to customers. Any uncalibrated index must be displayed as a non-probabilistic integer score.
- Do Not Transfer Association Football Models to American Football:** Do not adapt Dixon-Coles bivariate Poisson models, zero-inflation parameters, or goal-timing processes to NFL or college football scoring. American football scoring exhibits discrete, high-variance boundary conditions that Poisson processes fail to model.
- Do Not Reopen Hawkes Processes as Momentum Signals:** The Hawkes process investigation demonstrated a self-excitation parameter of $\hat{\alpha} = 0$, rejecting the presence of measurable in-game momentum.
- Do Not Reintroduce Features W5–W8:** Wasserstein distance, Detrended Fluctuation Analysis, Intrinsic Dimension, and Permutation Entropy failed pre-registered empirical trials across 12 out of 12 test fixtures. They are permanently removed from feature

stores.

- **Do Not Deploy Venn-Abers Corrections Over Inverted Metrics:** Venn-Abers calibration guarantees valid probability intervals over calibrated nonconformity scores, but it cannot fix fundamentally misranked, anti-predictive scoring metrics.
- **Do Not Deploy Adaptive Conformal Inference if Window Widths Fluctuate Widely:** If early-season uncertainty intervals expand by more than 50% relative to stable historical distributions, disable adaptive conformal inference and fall back to static historical calibration quantiles.
- **Do Not Quote Narrative Features Without Empirical Measurement:** Features such as unverified player equipment adjustments, generalized fourth-down heuristics, or untracked training camp narratives must not be incorporated into pricing engines.
- **Do Not Bypass the Early-Season EPA Data Floor:** Enforce `NFL_EPA_MIN_GAMES=4` strictly. Computing opponent-adjusted EPA from small early-season samples introduces substantial estimation error into team ratings.
- **Do Not Expand Model Scope into New Sports Prematurely:** Prohibit engineering work on NBA, NHL, or soccer models until NFL, NCAA, and MLB pipelines demonstrate full calibration and operational stability.
- **Do Not Market Platform Features as Artificial Intelligence:** Prohibit the use of vague marketing terms like "AI" or "deep neural network edge" in documentation, UI copy, and marketing materials. GSE is a sports analytics platform grounded in probability theory and empirical physics.
- **Do Not Treat Early-Season Game Samples as Season-Long Indicators:** The 17 games completed in Weeks 1 and 2 of 2026 provide an insufficient sample for updating macro-level parameters. Prior parameters must remain anchored to the historical 14,251 fixture corpus.
- **Do Not Use Small Live Settlement Samples to Claim Statistical Calibration:** Seventy live settled games cannot establish statistical calibration (≥ 1,000 required). Historical backtest records must be transparently incorporated into public calibration reporting.

H. The Recomputable Standard: What "100" Looks Like

The Formal Audit Statement

When Galaxy Sports Edge achieves a readiness score of 100, its public methodology statement will be verifiable by third-party auditors and quantitative market participants:

"Galaxy Sports Edge provides real-time, distribution-free sports forecasting calibrated across 14,251 decided NFL fixtures (1999–2025) and verified against live market prices. Over ≥ 1,000 production selections, its observed outcome frequencies match forecast true probabilities with an Expected Calibration Error below 0.025 and a Brier score superior to closing sportsbook consensus (< 0.024). Its marginal 90% prediction intervals maintain verified coverage of 0.90 ± 0.015 via non-exchangeable conformal inference. Across all published recommendations, it documents a positive Closing Line Value beat-rate exceeding

52.4% under immutable, timestamped accounting. All underlying datasets, execution logs, and validation code are publicly executable via open cryptographic receipts without proprietary dependencies."

Public Independent Audit Pipeline

The platform enforces public verifiability through an open four-stage verification architecture that decouples mathematical validation from internal application state.

1. Cryptographic Ledger ([/performance/audit](#))

Every published pick generates an immutable, append-only record secured via SHA-256 hash chaining:

$$\text{Receipt}_t = \text{SHA256}(\text{PickID}_t \parallel \text{Timestamp}_t \parallel \text{GameID}_t \parallel \text{OddsSnapshot}_t \parallel \text{ModelProb}_t \parallel \text{Receipt}_{t-1})$$

Timestamps are certified through an external RFC-3161 compliant timestamping authority to establish unalterable proof of publication prior to event start times.

2. Standardized Audit Inputs

Public audit records provide clean, tabular event vectors:

- **Selection Vector:** Timestamp, Sport, GameID, BetType, Selection, PublishedLine, ModelProb, ConformalLower, ConformalUpper.
- **Closing Market Vector:** Final consensus snapshot captured at kickoff (Pinnacle and Circa midpoints), VigFreeFairProb, FinalMarketLine.
- **Realized Outcome Vector:** Official league-settled final scores, box scores, player statistics, and settlement timestamps.

3. Automated Recomputation Engine

Auditors verify platform metrics by executing standardized mathematical calculations:

$$\text{ECE} = \sum_{m=1}^M \frac{\|B_m\|}{N} \left\| \text{acc}(B_m) - \text{conf}(B_m) \right\|$$
$$\text{Brier Score} = \frac{1}{N} \sum_{t=1}^N (P_t - Y_t)^2$$
$$\text{Conformal Empirical Coverage} = \frac{1}{N} \sum_{t=1}^N \mathbf{1}_{Y_t \in [L(X_t), U(X_t)]}$$
$$\text{CLV}_{\text{Decided}} = \frac{\sum_{t=1}^N \mathbf{1}_{\text{decided}}}{N_{\text{decided}}} \mathbf{1}_{\text{Beats Close}}$$
$$\text{CLV}_{\text{System}} = \frac{\sum_{t=1}^N \mathbf{1}_{\text{total}}}{N_{\text{total}}} \mathbf{1}_{\text{Beats Close}}$$

4. Audit Execution Script

A self-contained Python script is provided to enable public verification without internal

dependencies:

In-Scope vs. Out-of-Scope Production Boundaries

To maintain rigorous calibration standards, platform scope is strictly partitioned between certified production offerings and uncertified markets:

- **In-Scope Production Markets (Phase 4):**
 - *NFL Mainlines:* Pre-game point spreads, moneylines, and totals.
 - *NFL Player Props:* Passing touchdowns, receptions, and passing yards parameterized via Negative Binomial distributions.
 - *MLB Mainlines:* Pre-game moneylines and totals.
 - *NCAAF Mainlines:* Division I FBS point spreads and totals (restricted to Week 5 onwards, conditional on verified data pipelines).
- **Out-of-Scope Markets (Explicitly Prohibited):**
 - *In-Play / Live Betting:* Live betting markets across all sports.
 - *Same-Game Parlays (SGPs):* Multi-leg correlated parlays marketed as positive-expectation bets.
 - *Professional Basketball & Hockey:* NBA, WNBA, and NHL markets (prohibited until primary sports meet institutional standards).
 - *Association Football:* MLS, EPL, and European soccer leagues.
 - *Lower-Division Football:* FCS football, Division II/III, and spring developmental leagues.
 - *Unquantified Subjective Props:* Anytime touchdown scorers, defensive player props, and novelty event markets.

Failure Thresholds for Institutional Certification

The public auditing interface enforces binary pass/fail standards. Exceeding any of the following error thresholds results in immediate suspension of platform certification badges:

- **Brier Score Degradation:** If moneyline predictions yield a Brier score exceeding 0.204 over $n \geq 1,000$ decisions, the platform's "Market-Beating" status is revoked.
- **Calibration Error Expansion:** If Expected Calibration Error exceeds 0.025 across decile bins, pick generation is automatically disabled.
- **Conformal Coverage Breach:** If marginal coverage over rolling 500-pick windows drops below 88.5% or exceeds 91.5% at nominal $\alpha = 0.10$, prediction bands are flagged as uncalibrated.
- **Cryptographic Mismatch:** If independent recomputation of receipt hashes detects discrepancies with published ledger states, platform reporting is frozen pending security and data audit.